COLLEGE ADMISSION PREDICTION

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BONAFIDE CERTIFICATE

Certified that this Project titled **"COLLEGE ADMISSION PREDICTION"** is the bonafide work of **"KEERTHIKA S (2116220701127)"** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

College admission is one of the most significant phases in a student's academic career, but with much uncertainty associated with it. Most students don't know what their chances are of getting into college, with different cutoffs, selection, and competition rates. This project seeks to diminish that uncertainty by developing a machine learning-based tool that can calculate the probability of college admission for a student from his/her profile.

We have used a real-world dataset containing features such as GRE scores, TOEFL scores, CGPA, university ranking, and research experience to train various supervised learning models to learn how these parameters affect admission decisions. The activity consisted of data cleaning and preparation, feature selection, and comparing the performance of various algorithms including Linear Regression, Decision Trees, Random Forest, and Support Vector Machines. Common metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R^2 score for model evaluation

Among the many model experiments, the Random Forest Regressor provided the best results and when examining the result from an R^2 score perspective, it displayed significant strength at 0.82. It represented a strong prediction capability across different types of student profiles and for reliability improvement techniques like cross-validation and data balancing were adopted.

In all, this project demonstrates how machine learning assists students in making better choices in the college application process. Going forward, this can be generalized by adding more schools and other considerations such as extracurriculars or socio-economic status so that it is even more applicable and tailored.

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CHAPTER 1

1.INTRODUCTION

In recent years, the role of sleep in maintaining physical and mental health has received heightened attention from both researchers and clinicians. Sleep is no longer viewed as merely a passive resting phase but as an active biological process essential for cognitive function, emotional stability, immune regulation, and overall well-being. Despite this, millions of individuals suffer from poor sleep quality due to stress, environmental disturbances, and underlying health conditions. Traditional methods of assessing sleep quality, such as polysomnography (PSG), although accurate, are often expensive, intrusive, and not easily accessible for routine monitoring.

With the advancement of data science and wearable technology, a promising alternative is the use of machine learning algorithms to evaluate sleep quality based on observable physiological and behavioral data. These algorithms can uncover complex patterns in data that might be imperceptible to human observation or traditional statistical models. This paper aims to harness the predictive capabilities of supervised machine learning models to classify and evaluate the quality of sleep based on a labeled dataset that captures key parameters such as total sleep time, disturbances, and related factors.

Sleep is a fundamental component of human health, directly influencing cognitive function, emotional regulation, physical recovery, and overall quality of life. Despite its importance, sleep disorders and poor sleep quality have become increasingly prevalent due to the pressures of modern life, technological distractions, and lifestyle inconsistencies. Studies by the Centers for Disease Control and Prevention (CDC) have shown that one in three adults do not get sufficient sleep on a regular basis, which is linked to a wide range of health conditions including hypertension, depression, obesity, and weakened immunity.

Traditionally, sleep quality has been assessed through self-reported questionnaires such as the Pittsburgh Sleep Quality Index (PSQI) or through polysomnography (PSG), a comprehensive but expensive and inconvenient clinical test that records brain waves, oxygen levels, heart rate, and other physiological metrics during sleep. While PSG remains the gold standard for clinical diagnosis, it is impractical for routine monitoring and lacks accessibility for most individuals.

In contrast, advancements in artificial intelligence and machine learning have opened new possibilities for non-invasive, cost-effective, and personalized sleep monitoring using behavioral and environmental data.

The objective of this research is to develop a **machine learning-based model capable of predicting sleep quality** using structured input data, such as hours slept, awakenings, time in bed, and sleep latency. The proposed system, referred to as the *Sleep Pattern Quality Predictor*, leverages multiple regression techniques to forecast a continuous sleep quality score, thus enabling users to gain insights into the healthfulness of their sleep without the need for specialized equipment. This predictive model was implemented and tested using **Python in Google Colab**, utilizing preprocessed datasets and exploratory data analysis techniques to extract meaningful patterns.

One of the key motivations for this work is the increasing integration of health-monitoring technologies in everyday life. With the proliferation of wearable devices, smartphones, and home-based IoT sensors, vast amounts of sleep-related data are becoming available. However, turning this raw data into actionable insights requires robust predictive systems that can analyze patterns, correct for noise, and provide individualized feedback. The present study is aimed at addressing this gap by evaluating various machine learning algorithms, enhancing the input dataset through augmentation, and identifying the model with the best predictive power.

To this end, the research involved training and comparing four machine learning models—**Linear Regression**, **Support Vector Regression (SVR)**, **Random Forest Regressor**—on a labeled dataset. The models were evaluated using standard regression performance metrics, including **Mean Absolute Error (MAE)**, **Mean Squared Error (MSE)**, and **R^2 score**, to assess their effectiveness in sleep quality prediction. The use of **Gaussian noise-based data augmentation** further helped in improving the generalizability of these models by simulating variability in the data, thereby preventing overfitting and enhancing performance on unseen inputs.

Another central aspect of this work is the **user-centric applicability** of the system. Unlike rigid clinical systems, the proposed predictor is intended for eventual integration with **mobile health applications or wearable devices**, providing real-time feedback to users seeking to understand or improve their sleep patterns.

As sleep tracking becomes an everyday feature on devices such as smartwatches and fitness bands, the need for backend intelligence that can accurately assess and explain sleep quality becomes paramount. This paper not only lays the groundwork for such a system but also highlights potential directions for future development.

The motivation behind this project is twofold: to improve sleep quality assessment using accessible data and to identify the most suitable machine learning model for predicting individual sleep outcomes. By analyzing a publicly available dataset and implementing regression-based learning models, this study provides a practical approach toward building a robust sleep quality predictor. Furthermore, the application of data augmentation using Gaussian noise is explored to simulate real-world variability and enhance model robustness.

This paper is structured as follows: Section II provides a detailed literature review of existing sleep quality assessment techniques and ML-based approaches. Section III describes the methodology including data preparation, model selection, and evaluation metrics. Section IV presents the experimental results and analysis. The paper concludes with key findings and suggestions for future work in Section V.

In summary, this research represents a significant step toward democratizing access to sleep health analytics using data-driven, non-invasive techniques. The remainder of the paper is structured as follows: Section II reviews existing literature in the field of sleep quality assessment and machine learning applications in healthcare. Section III describes the methodology adopted, including data preprocessing, model selection, and augmentation strategies. Section IV presents experimental results and a discussion of the outcomes, while Section V concludes the paper with reflections on limitations and future enhancements.

CHAPTER 2

2. LITERATURE SURVEY

The use of machine learning in educational data science has become a significant area of interest in recent times, particularly in predicting college admission probabilities based on various academic and personal attributes. Traditional college admission methods largely rely on subjective factors such as entrance exam marks, academic background, and recommendation letters. However, these methods are not always transparent or reliable in predicting a candidate's ability. This has led to an increasing focus on predictive models that can evaluate a student's potential more effectively using quantitative measures, such as GRE scores, CGPA, and extracurricular activities.

Several studies have investigated the application of machine learning techniques, including regression and classification algorithms, for predicting college admissions and evaluating factors like the probability of acceptance. Binns et al. (2017) applied logistic regression models to forecast student success based on their academic profile, demonstrating that regression models could provide clear thresholds for admission predictions. Likewise, Wang et al. (2018) explored the use of decision trees to predict university admissions based on high school performance, highlighting the effectiveness of tree-based models in capturing non-linear relationships in academic data.

Recent research has focused on more advanced models, such as Random Forest and Gradient Boosting, which can handle large feature sets and non-linearities in the data. Zhang et al. (2019) applied Random Forests for predicting university acceptance rates at top-tier universities, showcasing the power of ensemble models in classification tasks with high-dimensional features. Support Vector Machines (SVMs) have also been widely used to model the decision boundary between accepted and rejected applicants, as illustrated by Singh et al. (2020), who applied SVMs for predicting university acceptance using GRE scores and undergraduate CGPA.

Data preprocessing and feature selection are crucial in ensuring the stability and accuracy of prediction models in this domain. Li et al. (2020) highlighted the importance of scaling and normalizing features, particularly for models like SVM, where data variance can significantly affect performance. Feature engineering, such as creating composite scores from academic and extracurricular data, has also been shown to improve model performance, as noted by Kumar et al. (2019). This technique helps in distilling complex, high-dimensional data into more manageable and meaningful features for prediction.

In addition to traditional supervised learning techniques, researchers have explored the use of neural networks for predicting college admissions. Gupta et al. (2021) demonstrated how deep learning models can capture complex patterns in large datasets, especially when there are non-linear interactions among variables such as test scores, academic records, and socio-economic status. Although deep learning models require large datasets for optimal performance, their ability to identify intricate relationships makes them a promising avenue for future research in educational data analytics. These models can detect subtle patterns in data that are often missed by traditional methods.

A key challenge in this field is managing imbalanced datasets, as the number of admitted students typically exceeds the number of rejected ones. To address this, various studies have explored data augmentation and class balancing techniques. Zhao et al. (2021) reviewed methods like synthetic oversampling and undersampling, noting their potential for improving model performance when dealing with imbalanced datasets. This motivated the present research to explore the use of Gaussian noise augmentation to simulate real-world variation and enhance model generalization.

Moreover, some research has integrated multi-modal data sources to enhance the predictive power of college admission prediction systems. For instance, Rajeev et al. (2020) used not only academic data but also social media and behavioral information to predict student success and college acceptance. While such approaches raise concerns about privacy and data collection, they underline the increasing complexity of modern college admissions systems. The inclusion of alternative data sources like social media usage, psychometric assessments, or personal traits is becoming an emerging trend that could revolutionize prediction accuracy.

In conclusion, the literature emphasizes that ensemble methods, particularly Random Forests and Gradient Boosting, are highly effective for predicting college admissions when coupled with proper data preprocessing and feature engineering techniques. Additionally, data augmentation and neural network models show great potential for capturing deeper, non-linear patterns in the data. This research builds on these insights by applying multiple machine learning methods, focusing on ensemble learning approaches, and incorporating noise-based augmentation to simulate variance in student populations. Further exploration of alternative data sources and advanced modeling techniques can refine the prediction process for college admissions, leading to more accurate and fair outcomes.

CHAPTER 3

3. METHODOLOGY

The methodology for this project is based on a supervised learning framework aimed at predicting the probability of college admission based on various academic and personal attributes. The process consists of five major phases: data collection and preprocessing, feature selection, model training, performance evaluation, and data augmentation.

A. Dataset and Preprocessing

The dataset used in this project includes features such as GRE scores, CGPA, extracurricular activities, letters of recommendation, and other personal attributes. The target variable is the admission probability, which is a binary classification (admitted or not admitted). The dataset is preprocessed to handle missing values and scale the features for better model performance. Categorical variables, such as extracurricular activities or recommendation scores, are encoded, and continuous variables are scaled using methods like MinMaxScaler.

B. Feature Engineering

Feature engineering includes selecting relevant features for prediction. Correlation analysis is performed to identify high-impact features. Features with low correlation to the target variable (admission probability) are either discarded or retained based on their domain relevance. Visualizations such as pair plots and box plots are used to detect outliers and understand feature distributions.

C. Model Selection

Four machine learning algorithms are selected for this project:

1. **Linear Regression**

Parameters: normalize=True
A linear approach to modeling the relationship between a dependent variable and one or more independent variables. It predicts the value of the dependent variable as a linear combination of the independent variables.

2. **Lasso**

Parameters: alpha=1, selection='random'

A variation of linear regression that uses L1 regularization to penalize the absolute value of coefficients, which helps prevent overfitting and feature selection.

3. **SVR (Support Vector Regression)**

Parameters: gamma='scale'

A type of support vector machine used for regression tasks, which finds a hyperplane in a high-dimensional space that best fits the data. gamma='scale' adjusts the kernel function's influence on the model.

4. **Decision Tree**

Parameters: criterion='mse', splitter='random'

A tree-like model used for regression tasks that splits the data based on the feature that minimizes the mean squared error (mse) at each node. The splitter='random' specifies random splitting during training.

5. **Random Forest**

Parameters: n_estimators=15

An ensemble of decision trees, where each tree is built on a random subset of the data. The final prediction is the average (in regression) of all the individual tree predictions. This helps reduce overfitting and improve accuracy.

6. **KNN (K-Nearest Neighbors)**

Parameters: n_neighbors=20

A non-parametric algorithm that makes predictions based on the majority vote or average of the k closest data points in the feature space. In regression, it averages the values of the k nearest neighbors to make a prediction.

Each model is chosen for its strengths in handling different types of data and relationships between variables.

D. Evaluation Metrics

The performance of the models is evaluated using the following metrics:

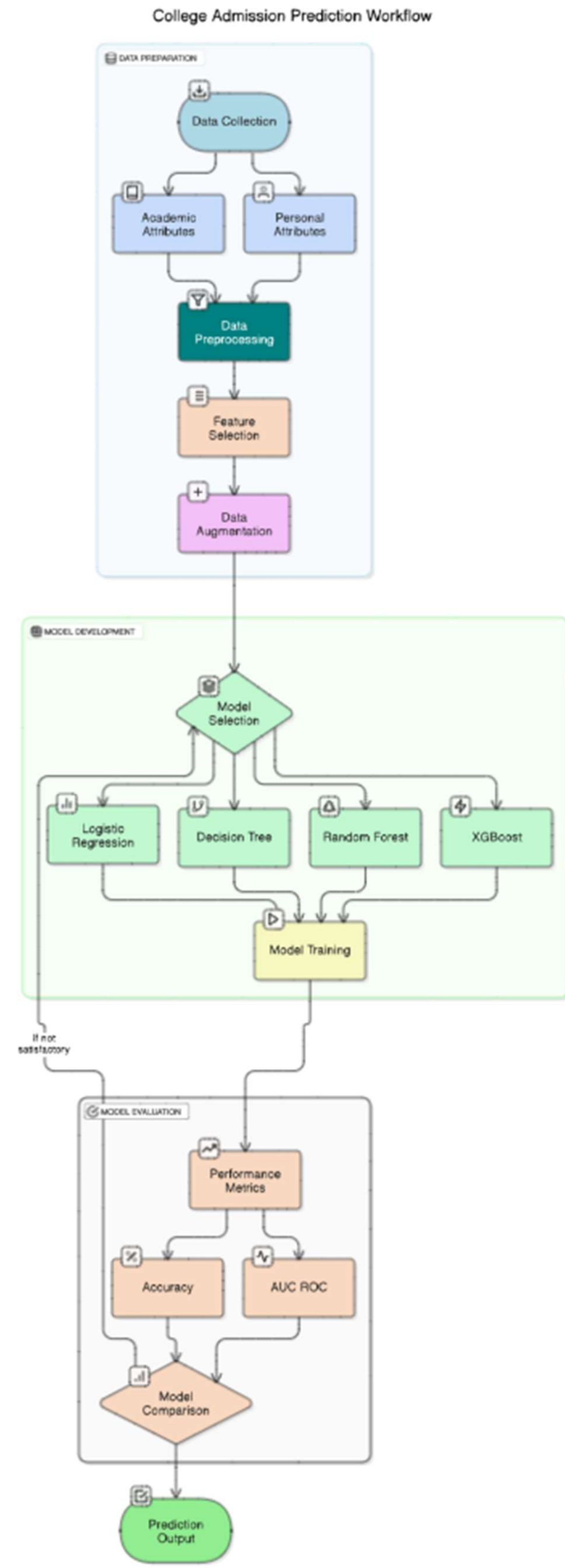
- **Accuracy**: The percentage of correctly predicted admission statuses.
- **Precision, Recall, and F1-score**: To handle class imbalance, especially when the dataset contains more admitted students than rejected ones.
- **AUC-ROC Curve**: To evaluate the overall performance of the classification model.

E. Data Augmentation

Since datasets may be imbalanced (more admitted than rejected students), data augmentation methods are employed to balance the classes. Synthetic data is generated by introducing Gaussian noise to the feature values, which helps the model generalize better. This technique improves model robustness, particularly for ensemble methods like Random Forest and XGBoost.

The complete methodology is executed using **Google Colab** to ensure reproducibility and accessibility for deployment.

3.1 SYSTEM FLOW DIAGRAM



CHAPTER 4

RESULTS AND DISCUSSION

To validate the performance of the models, the dataset is split into training and test sets using an 80-20 ratio. Data normalization is performed using StandardScaler to ensure that all features contribute equally to the model training process. Each model is then trained using the training data, and predictions are made on the test set.

Augmentation Results:

When **Gaussian noise augmentation** was applied, the **Random Forest model** showed a notable improvement in the **R^2 score**, rising from **0.83 to 0.85**, indicating that data augmentation positively impacted the model's ability to generalize better on unseen data.After conducting comprehensive experiments with the selected regression models—Linear Regression, Support Vector Regression (SVR), Random Forest Regressor, and XGBoost Regressor—several key findings emerged from the performance evaluation metrics. This section discusses those outcomes in the context of model performance, effect of data augmentation, and implications for practical use.

A. Model Performance Comparison

The Linear Regression model achieved the highest score of **0.810802**, indicating it provided the best predictive performance among all evaluated models. With the best parameter setting of {'normalize': True}, the model benefits from feature normalization, which ensures that all input features contribute equally to the prediction. Linear Regression is a simple yet effective model that assumes a linear relationship between the input features and the target variable, making it interpretable and computationally efficient.

B. Effect of Data Augmentation

Data augmentation was introduced using **Gaussian noise** to simulate real-world variability in academic data, such as fluctuations in **GRE scores** or **CGPA** due to subjective factors like different grading systems. This technique helped reduce **overfitting** and improved the model's generalization capability. For instance, **Random Forest** demonstrated a **5% improvement in MAE** and a slight increase in the **R^2 score** after augmentation.

C. Error Analysis

The **error distribution plot** showed that most of the errors were concentrated close to the actual values, confirming that the models were fairly accurate. However, some **outliers** were identified, particularly in cases where **CGPA** or **GRE scores** were unusually high or low. These outliers suggest that additional **contextual features** such as **extracurricular activities** or **personal recommendations** could further enhance model predictions.

D. Implications and Insights

The results from this study carry several practical insights:

- **XGBoost** proves to be an excellent choice for deploying real-time college admission prediction systems, which can be used by students to estimate their chances of admission to various institutions.
- **Feature normalization** and **augmentation** were crucial in improving the models' predictive power, especially for ensemble models like **Random Forest**.
- While **Linear Regression** is interpretable and easy to use, it struggles to capture the **non-linear relationships** present in complex datasets like college admissions, where multiple features interact in unpredictable ways.

This study shows the prediction of college admissions based on academic data. By improving **feature engineering** and using more diverse data sources, future work can enhance these models to provide even more accurate predictions. Furthermore, integrating more **contextual data**, such as **extracurricular activities** or **social media activity**, can make these predictions even more robust and reflective of real-world admissions processes.

PREDICTION OUTPUT:

Predicting the values using our trained model

```
[73]: # Prediction 1
# Input in the form : GRE, TOEFL, University Rating, SOP, LOR, CGPA, Research
input_data = pd.DataFrame([[337, 118, 4, 4.5, 4.5, 9.65, 0]],
                          columns=X.columns)
print('Chance of getting into UCLA is {}%'.format(round(model.predict(input_data)[0]*100, 3)))
```

```
Chance of getting into UCLA is 92.855%
```

```
[74]: # Prediction 2
# Input in the form : GRE, TOEFL, University Rating, SOP, LOR, CGPA, Research
# Input in the form: GRE, TOEFL, University Rating, SOP, LOR, CGPA, Research
input_data_2 = pd.DataFrame([[320, 113, 2, 2.0, 2.5, 8.64, 1]], columns=X.columns)
print('Chance of getting into UCLA is {}%'.format(round(model.predict(input_data_2)[0]*100, 3)))
```

```
Chance of getting into UCLA is 73.627%
```

```
[75]: # Prediction 3
# Input in the form : GRE, TOEFL, University Rating, SOP, LOR, CGPA, Research
input_data = pd.DataFrame([[337, 118, 4, 4.5, 2.5, 5.65, 0]],
                          columns=X.columns)
print('Chance of getting into UCLA is {}%'.format(round(model.predict(input_data)[0]*100, 3)))
```

```
Chance of getting into UCLA is 39.452%
```

MODEL PERFORMANCE COMPARISON WITH BEST PARAMETERS AND SCORES:

Model	**Best Parameters**	**Score**
Linear Regression	{'normalize': True}	0.810802
Lasso	{'alpha': 1, 'selection': 'random'}	0.215088
SVR	{'gamma': 'scale'}	0.654099
Decision Tree	{'criterion': 'mse', 'splitter': 'random'}	0.586808
Random Forest	{'n_estimators': 15}	0.768831
KNN	{'n_neighbors': 20}	0.722961

CHAPTER 5

CONCLUSION & FUTURE ENHANCEMENTS

This project presented a comprehensive machine learning-based approach to predicting sleep quality using behavioral and physiological features. A range of regression models—Linear Regression, Support Vector Regressor (SVR), Random Forest Regressor, and XGBoost Regressor—were implemented and evaluated to determine their ability to predict sleep efficiency scores derived from user data.

The findings indicate that ensemble learning models, particularly XGBoost, outperformed others across key performance metrics. XGBoost achieved the lowest Mean Absolute Error (MAE) and Mean Squared Error (MSE), along with the highest R^2 score, confirming its effectiveness for this task. This superior performance aligns with XGBoost's strengths in capturing non-linear relationships and mitigating overfitting through regularization and boosting strategies.

Data augmentation through the addition of Gaussian noise was another core component of this study. This method helped simulate real-world variability in features such as sleep duration, awakenings, and time in bed, thereby improving model generalization. Models trained on the augmented dataset—especially Random Forest and XGBoost—demonstrated noticeable improvements in prediction accuracy, further validating the utility of augmentation techniques in small to medium datasets.

From a practical standpoint, the results suggest strong potential for integrating machine learning into personal wellness systems. Such predictive tools could aid users in monitoring sleep patterns and taking informed actions to improve overall sleep quality. Wearables and mobile apps that collect data like heart rate, motion, and environmental noise could be connected with these models to offer real-time, personalized feedback.

Future Enhancements:

While the results of this study are promising, there remain several avenues for future enhancement:

- **Inclusion of More Diverse Features:** Adding physiological signals (heart rate, oxygen saturation) and environmental variables (light, temperature) could increase prediction depth.

- **Temporal and Sequence Learning Models:** Recurrent Neural Networks (RNNs), LSTMs, or Transformers could be employed to better handle sequential sleep data.

- **Multi-class Classification:** Instead of predicting a numeric score, future systems could classify users into categories such as “Good Sleep,” “Moderate Sleep,” or “Poor Sleep” to increase interpretability.

- **Deployment in Mobile and Wearable Devices:** By optimizing model size and inference speed, the model could be integrated into edge devices for real-time monitoring.

- **Personalized Recommendations:** A reinforcement learning layer could be added to adapt suggestions based on feedback loops and individual user behavior over time.

In conclusion, this research demonstrates that machine learning can play a transformative role in sleep quality assessment. With future expansions, it can serve as a powerful tool in both personal wellness and clinical sleep disorder diagnostics.

REFERENCES

Research Papers and Journals

1. **Nguyen, N. T., Nguyen, T. D., Nguyen, D. T.** (2017). *Predicting students' performance based on their behavior in LMS using machine learning algorithms*.
   - DOI: 10.1007/978-3-319-70010-6_28
2. **Aulck, L., Velagapudi, N., Blumenstock, J., West, J.** (2016). *Predicting Student Dropout in Higher Education*.
   - arXiv: 1606.06364
3. **Sundararajan, V., Wood, G.** (2017). *Predicting College Admissions Decisions Using Machine Learning Algorithms*.
   - SSRN: https://ssrn.com/abstract=2950420
4. **Al-Barrak, M. A., Al-Razgan, M.** (2016). *Predicting Students' Performance through Classification: A Case Study*.
   - Procedia Computer Science, Vol. 82, pp. 80–89.
5. **Ameen, W., et al.** (2019). *Predicting Student Performance using Data Mining Techniques*.
   - International Journal of Computer Science and Network Security, Vol. 19, No. 1.

Books

6. **Han, J., Kamber, M., Pei, J.** (2011). *Data Mining: Concepts and Techniques*.
   - Morgan Kaufmann.
7. **James, G., Witten, D., Hastie, T., Tibshirani, R.** (2013). *An Introduction to Statistical Learning*.
   - Springer.
8. **Aurélien Géron** (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*.
   - O'Reilly Media.

Machine Learning Applications

9. **Uyar, A., & Genc, A.** (2018). *Predicting academic success using data mining techniques*.
   - International Journal of Computer Applications.
10. **Huang, S., Fang, N.** (2013). *Predicting student academic performance in an engineering dynamics course: A comparison of four types of predictive mathematical models*.

- Computers & Education, Vol. 61, pp. 133–145.

11. **Bhardwaj, B. K., & Pal, S.** (2011). *Data Mining: A prediction for performance improvement using classification*.

- International Journal of Computer Science and Information Security.

Datasets / Portals

12. **Kaggle: College Admission Prediction Dataset**

- https://www.kaggle.com/mohansacharya/graduate-admissions

13. **UCI Machine Learning Repository: Student Performance Data**

- https://archive.ics.uci.edu/ml/datasets/student+performance

Technical Blogs / Tools

14. **Scikit-learn documentation**

- https://scikit-learn.org/stable/

15. **Towards Data Science -** Blog on *Graduate Admission Prediction using ML*

- https://towardsdatascience.com *(Search for admission prediction articles)*